Robust Drug-Target Affinity Prediction Under Distribution Shift: Uncertainty-Aware Ensemble Learning for the STAR AI Competition

Shikhar Verma
Georgia Institute of Technology
sverma87@gatech.edu

Ninh Le
Georgia Institute of Technology
nba9@gatech.edu

1 Introduction

Predicting drug-target binding affinity (DTA) is fundamental to computational drug discovery, enabling the prioritization of candidate compounds for expensive wet-lab validation. While deep learning models have shown strong performance on curated DTA benchmarks [1, 2], real-world deployment introduces distribution shifts that silently degrade predictions: novel chemical scaffolds absent from training data, and measurements from assay types with different noise characteristics.

This specific Track 2 competition operationalizes this challenge through three test splits of increasing difficulty: Split A (random, in-distribution), Split B (scaffold shift), and Split C (assay shift). The primary ranking metric is RMSE, while the offline Shift-Robust Concordance Index (SR-CI $=0.30\times\text{CI}_A+0.35\times\text{CI}_B+0.35\times\text{CI}_C$) emphasizes generalization by upweighting the harder splits.

Our approach addresses these challenges through three key design decisions: (1) a more expressive GNN architecture that leverages bond-level features, (2) a robust loss function that handles label noise inherent in mixed-assay datasets, and (3) an uncertainty quantification framework that detects when predictions are unreliable due to distribution shift.

2 Methods

2.1 Data Harmonizer (Phase 1)

The training set comprises approximately 25,000 drug-target pairs with pKi labels. Each sample carries a split membership label indicating which test regimes (A, B, C) it participates in, with membership combinations including A∩B∩C (48.5%), A∩C (21.5%), B∩C (20.6%), and C-only (9.4%).

To prevent the model from over-fitting to the majority group, we employ inverse-frequency weighting via a `WeightedRandomSampler`, where samples from under-represented membership groups (e.g., C-only) receive proportionally higher sampling probability (up to $10.7\times$ for C-only). A 90/10 stratified split preserves group proportions in our internal validation set (2,578 samples).

2.2 Graph Encoder (Phase 2)

Each drug molecule is converted to a molecular graph with 29-dimensional atom features (element type, degree, formal charge, aromaticity, hybridization) and 7-dimensional bond features (bond type, conjugation, ring membership). We additionally compute 2,048-bit Morgan fingerprints (radius 2) as auxiliary features. Protein targets are encoded using mean-pooled ESM-2 embeddings (650M parameter model, 1,280-dimensional output).

2.3 Affinity Predictor (Phase 3)

Compound encoder. We use a 4-layer Graph Isomorphism Network with edge features (GINEConv) [3], which incorporates bond-level information via the `edge_dim` parameter. Each layer applies:

$$h_v^{(l+1)} = \text{MLP}^{(l)}\left((1+\epsilon^{(l)})\cdot h_v^{(l)} + \sum_{u\in\mathcal{N}(v)} \text{ReLU}(h_u^{(l)} + e_{uv}^{(l)})\right) \tag{1}$$

where $e_{uv}^{(l)}$ is the edge feature embedding, followed by BatchNorm, ReLU, and dropout (p=0.2). Graph-level readout uses mean pooling.

Auxiliary fingerprint branch. The 2,048-bit Morgan fingerprint is projected to 256 dimensions via a linear layer with ReLU and dropout. This provides a complementary global molecular descriptor that captures substructure presence without relying on the learned graph representation.

Protein encoder. Pre-computed ESM-2 embeddings (1,280-d) are projected to 256 dimensions through a two-layer MLP with dropout.

Prediction head. The compound graph embedding (256-d), fingerprint embedding (256-d), and protein embedding (256-d) are concatenated (768-d total) and passed through a 3-layer MLP ($768\to256\to128\to2$) with ReLU activations and dropout between layers, outputting the "mean" and "variance" of the affinity of each compound. The mean of the "means" predicted by models in the ensemble is used as the final affinity for downstream analyses, including competition submission.

2.4 Training Strategy

Loss function. We use heteroscedastic NLL loss rather than MSE. The first term acts as precision, down-weighting the MSE during backpropagation for samples the dual-output model identifies as highly noisy, while the second term acts as a regularizer preventing the model from predicting infinite uncertainty. We also employed gradient clipping to prevent explosions from extreme precision values.

$$\mathcal{L} = \frac{1}{2} e^{-\log(\sigma^2)} (y-\mu)^2 + \frac{1}{2} \log(\sigma^2) \tag{2}$$

Optimization. Adam optimizer (lr $= 10^{-3}$, weight decay $= 10^{-5}$) with CosineAnnealingWarmRestarts (T_0=20, T_{mult}=2, $\eta_{\min}$=10^{-6}). Gradient clipping at max norm 5.0. Model checkpointing on best validation **RMSE** (the Kaggle ranking metric). Early stopping with patience of 20 epochs.

Deep ensemble. We train 5 independently initialized models with different random seeds, providing prediction diversity. At inference, predictions are averaged across all members.

2.5 STAR Analysis (Phase 4)

Safety. Uncertainty estimation via MC Dropout and ensemble disagreement prevents over-confident screening recommendations. Rather than returning a single affinity prediction, our pipeline provides confidence intervals in the form of $\log$ variance, coupled with MC dropout, enabling practitioners to set safety thresholds before committing wet-lab resources to uncertain candidates. More specifically, during inference, dropout layers remain active across T=10 stochastic forward passes per ensemble member (50 total passes). In doing so we can compute both epistemic variance, which is the model ignorance uncertainty and computed as the variance of the 50 variances, and aleatoric variance, which is stochastic uncertainty and computed as mean of the variances. The total variance is the sum of the two and is used for every downstream analysis.

Trustworthiness. Shift-aware evaluation across splits A, B, and C explicitly exposes where model predictions are fragile. By computing per-split RMSE and CI, we make performance degradation visible rather than hiding it behind aggregate metrics (Table 1). The ensemble variance signal provides an interpretable indicator of *which specific predictions* are least reliable, grounding the model.

Actionability. The combined affinity prediction and uncertainty score enable budget-constrained compound ranking. In a real drug discovery setting, this translates to a ranked candidate list where high-affinity, low-uncertainty compounds are prioritized—maximizing hit rate under a fixed experimental budget. More specifically, candidates for wet lab testing are selected by (1) filtering for compounds with predicted variance less than some set max variance, (2) computing utility = predicted affinity $- \lambda_{\text{risk}} \cdot$ predicted std of the remaining compounds and (3) ranking them by this utility metric. The more risk-averse you are (for example when the assays are expensive), the higher λ_{risk} should be set (default at 1 but can be raised to 2 or 3).

Responsibility. We transparently report performance degradation across splits and scaffold novelty. By publishing per-split metrics alongside overall RMSE, we avoid overstating model capabilities. For our model, RMSE understandably degrades from 0.56 on Split A to 0.59 on Split B, with average uncertanty also increasing from 0.58 to 0.61 (Table 1). Note that the evaluation metrics were all computed on our validation test given unless stated otherwise.

Table 1: Shift-Aware Degradation

Split	N	RMSE	Avg. Uncertainty
A (Random)	1,804	0.558697	0.567327
B (Scaffold)	1,782	0.592485	0.614165
C (Assay)	2,578	0.558007	0.568717

3 Ablations

3.1 Ablation 1: Ensemble Size

We wanted to see how decreasing the number of total models in our 5-model ensemble affected the RMSE. When we removed models from the original, we found a decrease in the total variance of

the predictions. Furthermore, averaging more models allowed the final prediction to stabilize and generalize more to the testing set, enabling a lower RMSE value.

Table 2: Effect of ensemble size on prediction quality.

Inference Strategy	**Val RMSE ↓**	**Passes**
Single model (deterministic)	0.6633	1
3-model ensemble	0.5609	3
5-model ensemble	0.5411	5

3.2 Ablation 2: Distribution Shift Analysis

3.2.1 Balancing Groups during Training Time

The training data exihibits significant class imbalance across the split membership groups: A∩B∩C comprises 48.5% of samples, while the C-only group (assays) accounts for 9.4%. Without adjusting for this, the model would disproportionately optimize for hte majority group at the expense of under-represented regimes. We address through inverse-frequency sampling weights computed in Phase 1, where the C-only samples receive $10.7\times$ the sampling probability of A∩B∩C samples. This was implemented through Pytorch's `WeightedRandomSampler`, which rebalances each training epoch so that ever split membership group contributes equally to the gradients.

3.2.2 Dual-Metric Evaluation

Althouh we checkpoint models based on RMSE, we also compute the Shift-Robust Concordance Index (SR-CI), utilizing the formula SR-CI $= 0.30 \times \text{CI}_A + 0.35 \times \text{CI}_B + 0.35 \times \text{CI}_C$ at every epoch during training. This dual metric allows us to accomplish 2 things:

1. Provide an early signal if RMSE improvements come at the cost of ranking quality on the tougher splits
2. Ensure we can report SR-CI in the methodology assessment without re-running the evaluation multiple times

Table 3: Per-model training results. All models were checkpointed on best validation RMSE. SR-CI is tracked but not used for model selection.

Model	**Best Val RMSE ↓**	**Best Val SR-CI ↑**	**Checkpoint Epoch**	**Note**
Model 1	0.5658	0.8653	97	
Model 2	0.5624	0.8623	100	
Model 3	0.5783	0.8592	99	
Model 4	0.6141	0.8452	53	Early convergence
Model 5	0.5531	0.8683	100	Best individual
Mean	**0.5747**	**0.8601**	—	

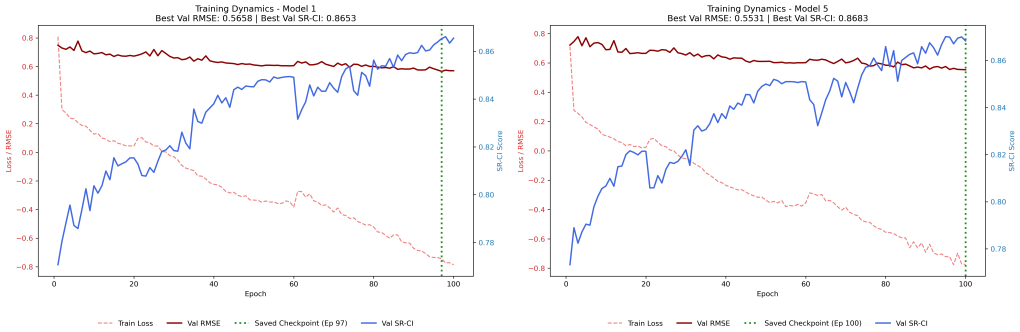

Figure 1: Training dynamics for Model 1 (left) and Model 5 (right, best individual). Validation RMSE (dark red, left axis) decreases steadily while SR-CI (blue, right axis) rises in tandem, confirming that RMSE optimization with balanced sampling implicitly improves shift-robust ranking. The green dashed line marks the saved checkpoint (best RMSE). All five models exhibit this correlated improvement pattern.

4 Reproducibility

Code and environment. All experiments use Python 3.10, PyTorch $\geq$2.1, PyTorch Geometric $\geq$2.4, and RDKit $\geq$2023.03. The complete pipeline runs in a series of jupyter notebooks with a Conda environment specification (`environment.yml`). Our GitHub link for the code can be found here.

Compute. Training was conducted on Georgia Tech PACE Phoenix and local hardware. The full pipeline (5-model ensemble training + OOD analysis + submission generation) completes in approximately 3 hours.

Hyperparameters. Hidden dimension: 256. GINEConv layers: 4. Dropout: 0.2. Learning rate: 10^{-3}. Weight decay: 10^{-5}. Batch size: 128. Early stopping patience: 20. Ensemble size: 5.

Data splits. 90/10 stratified train/validation split, stratified by split membership. Test set: 23,191 samples provided by competition organizers (no labels).

5 Final Evaluation Metrics

We utilized these four following metrics to evaluate our models:

(1) **Ranking Stability (Kendall's τ)** evaluates how correctly the model ranks the compound. More specifically, within each split (A, B and C) we computed Kendall's τ for the top 200 compounds predicted by our model as best binders and observed uniformly significant correlations across all splits (Table 4).

Table 4: Ranking Stability (Kendall's τ, top-200)

Split	N	Kendall's τ	p-value
A (Random)	1,804	0.2592	5.5e-08
B (Scaffold)	1,782	0.2633	3.43e-08
C (Assay)	2,578	0.2751	7.96e-09

Table 5: Hit Rate @ Budget (top-200 predictions)

Split	Hit Rate @ 200	Recall
A (Random)	0.910	0.396 (182/460 hits)
B (Scaffold)	0.920	0.352 (184/523 hits)
C (Assay)	0.930	0.287 (186/649 hits)

(3) **Calibration** Expected Calibration Error (ECE) checks if the model's confidence matches Gaussian distribution. Since our model uses MC Dropout, each predicted value has $\pm$ a standard deviation (σ), and theoretically 68.3% of true values should fall within $\pm 1\sigma$, and 95.4% within $\pm 2\sigma$, the deviation from which is computed by ECE. as many as 91% of our predicted value fall within 1σ, suggesting it is under-confident. However, an ECE value of 0.0843 is promising.

(4) **Abstention Quality (Precision)** evaluates the ability of our model to flag a compound as highly uncertain. Calculating the overlap between the model's top 25% uncertain predictions and its top 25% worst predictions, we observed that the model achieved a 69% precision on the abstention set.

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